RISK 201: Measurement of Interest

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1.1 Definitions

Principal (P):	The amount of money (capital) we start with
Accumulated value (S):	The amount of money at the end of a defined period of time, without adding/withdrawing any capital
Amount of interest (I):	The difference between the accumulated value and the principal

Accumulation Function

Let t measure time from the date of the original investment.

The unit of t is called the **measurement period** (usually days, months or years). The default is one year.

Definition

The **accumulation function**, $a(t)$, $t \geq 0$, gives the accumulated value at time t of an original investment of 1 unit of principal.

Accumulation Function, Examples









Accumulation Function, Properties

1. $a(0) = 1$
2. $a(t)$ is usually increasing in time
3. We consider $a(t)$ to be continuous

Example 1.1

You are given $a(t) = bt^2 + ct + d$. You are also told that $a(1) = 4.5$ and $a(2) = 9$. Find $a(5)$.

Amount Function

Definition

The **amount function**, $A(t)$, $t \geq 0$, gives the accumulated value at time t of an original investment of P units of principal.

Amount Function, Properties

1. $A(0) = P$
2. $A(t)$ is usually increasing in time
3. We consider $A(t)$ to be continuous

Example 1.2

Use the same accumulation function as in Example 1.1. If \$200 is invested at time 0, what is the accumulated value at time 7?

Interest Earned

The **amount of interest earned during the nth period** is given by

$$I_n = A(n) - A(n-1), \qquad n = 1, 2, \ldots$$

Example 1.2, continued

How much interest is earned between times 5 and 7 on our initial investment of \$200?

1.2 Effective Rate of Interest

Definition

The **effective rate of interest** is a measure of interest denoted by i and defined as the amount of money that 1 unit invested at the beginning of a period will earn during the period, where interest is paid at the **end** of the period.

$$i = \frac{a(1) - a(0)}{a(0)} = a(1) - a(0) \quad \text{(why?)}$$

Effective Rate of Interest, cont'd

The effective rate of interest earned in the nth period:

$$i_n = \frac{a(n) - a(n-1)}{a(n-1)}$$

Example 1.2, continued

Calculate the effective rates of interest i_1, i_2 and i_{10}.

1.3 Simple Interest

Simple interest refers to a specific way of accumulating interest:

$$a(t) = 1 + rt$$

where r is the annual rate of simple interest and $t \geq 0$ is time expressed in years.

Simple interest is often used for short-term transactions (less than a year).

Calculating t

The time period in question may be stated:

- In years ("a loan for half a year")
- In months ("an investment for 5 months")
- Using actual dates ("Mar 12 to Sep 25, 2022")

Calculating t, continued

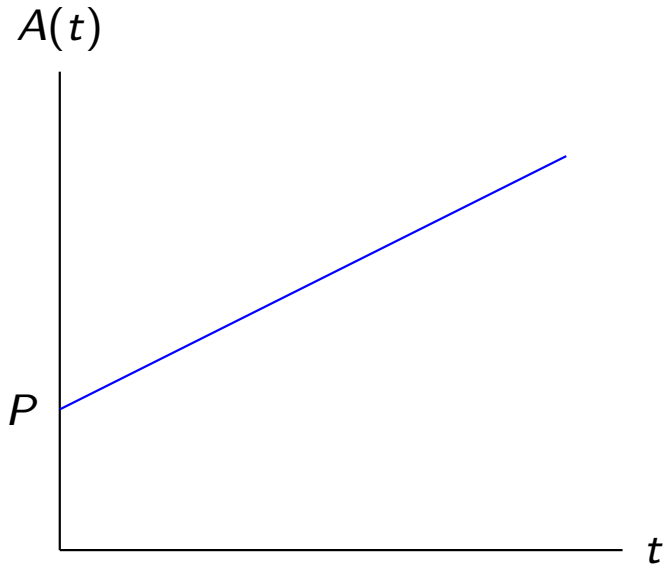
Three different conventions when dates are given:

- Actual/actual (exact interest)
- Actual/360 (Banker's rule)
- 30/360 (ordinary interest)

$$\text{numerator} = (D_2 - D_1) + 30(M_2 - M_1) + 360(Y_2 - Y_1)$$

$A(t)$ for simple interest

$$A(t) = P \cdot a(t) = P(1 + rt), \qquad t \geq 0$$



The amount of simple interest accumulated over a period of length t is

$$I = Prt$$

Example 1.3

You invest \$500 at 7% simple interest. What is the accumulated value of your investment after 30 days?

What is it after 2 years?

Effective rate under simple interest

With a constant rate of simple interest, r, the effective rate of interest during the nth period is

$$i_n = \frac{I_n}{A(n-1)}$$

1.4 Compound Interest

Compound interest is another common way of accumulating interest:

$$a(t) = (1 + i)^t$$

where i is the annual rate of compound interest and t is time.

The amount function is

$$A(t) = P(1 + i)^t$$

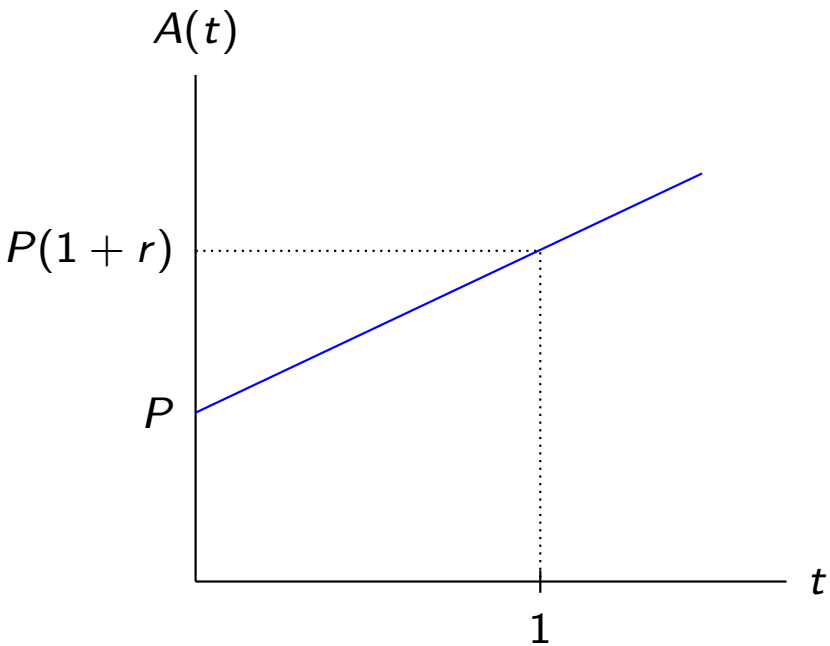
Example 1.4

You invest \$500 at 7% compound interest. What is the accumulated value of your investment after 3 months?

After 2 years?

Compare your answers to Example 1.3.

Simple vs. compound interest



With simple interest:

$$A(t) = P(1+rt)$$
$$A(0) = P$$
$$A(1) = P(1+r)$$

With compound interest:

$$A(t) = P(1+r)^t$$
$$A(0) =$$
$$A(1) =$$

Interest on interest

With compound interest, each interest payment is assumed to be automatically **reinvested**. There is then interest being earned not just on the principal but also on all previously accumulated interest.

$$
\begin{aligned}
a(1) &= 1 + i \\
a(2) &= (1+i)^2 = 1 + i + i(1+i) \\
&\vdots \\
a(t) &= (1+i)^t = (1+i)^{t-1}(1+i) \\
&= (1+i)^{t-1} + i \cdot (1+i)^{t-1}
\end{aligned}
$$

Effective rate under compound interest

With a constant rate of compound interest, i, the effective rate of interest during the nth period is

$$\begin{aligned} i_n &= \frac{A(n) - A(n-1)}{A(n-1)} \\ &= \frac{(1+i)^n}{(1+i)^{n-1}} - 1 \\ &= i \end{aligned}$$

Special property of compound interest

The accumulation function for compound interest satisfies the property

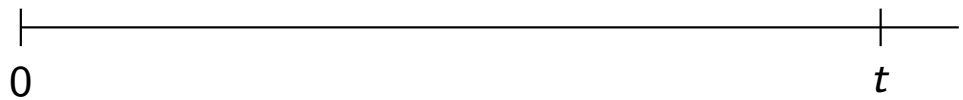
$$a(t+s) = a(t) \cdot a(s) \qquad \forall\, t \geq 0,\ s \geq 0$$



This is a useful property unique to compound interest.

1.5 Present Value

The accumulation function connects the value of an investment today to its value at some future date.



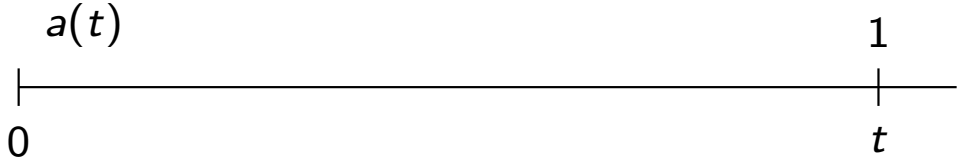
But what if we want to target a particular amount at a specific future date? How much should we set aside today?

Definition of present value

Definition

The **present value** (at time 0) of 1 due in t years is the amount needed at time 0 which will accumulate to 1 by time t.

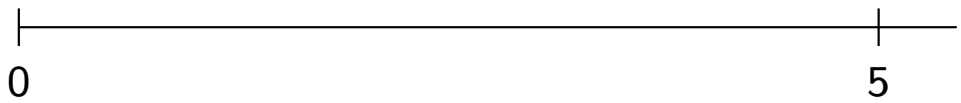
$$PV = \frac{1}{a(t)}$$



The present value (at time 0) of k due in t years is

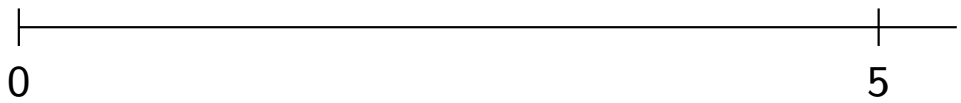
Example 1.5

(a) What is the present value of \$600 due in 5 years assuming **simple** interest at 10% per year?



Example 1.5, continued

(b) What is the present value of \$600 due in 5 years assuming **compound** interest at 10% per year?



Example 1.5, continued

(c) What is the present value of \$600 due in 5 years with accumulation function $a(t) = t^2 + 2t + 1$?

Discounting

The process of finding the PV of a payment, or stream of payments, is called **discounting**.

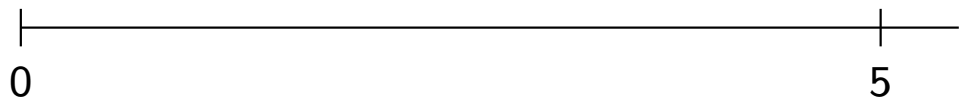
Discounting with compound interest involves multiplying by a factor $\frac{1}{(1+i)^t}$.

$$\text{Let } v = \frac{1}{1+i}.$$

This is the one-period present value factor, also known as the **discount factor**.

Properties

1. As the interest rate increases, the present value



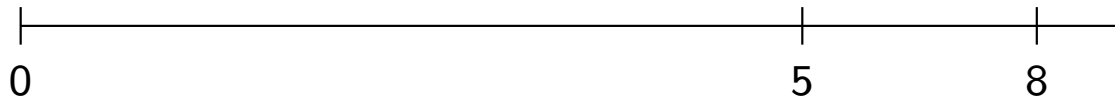
Properties, continued

Mathematically,

$$\frac{d}{di}PV = \frac{d}{di}\frac{k}{(1+i)^t} = k \cdot \frac{d}{di}(1+i)^{-t}$$

Properties, continued

2. As the time horizon increases, the present value



$$\frac{d}{dt}PV = \frac{d}{dt}\frac{k}{(1+i)^t} = \frac{-k\ln(1+i)}{(1+i)^t}$$

Key concepts from week 1

1. Money has **time value**: k today $\neq k$ any other day

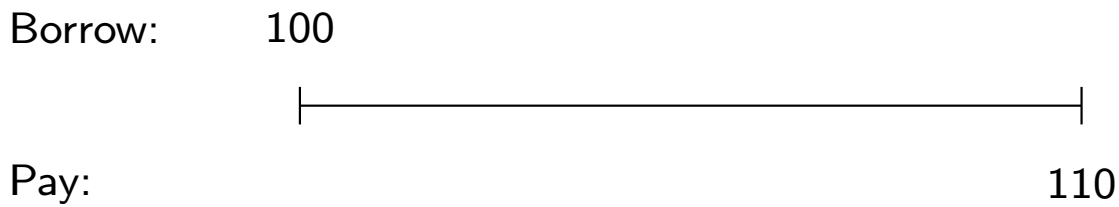
2. The **present value** tells us how much a cash flow occurring in the future is worth today.

3. The **effective rate of interest** measures the amount of interest as a proportion of the value at the beginning of the period.

1.6 Discount rates

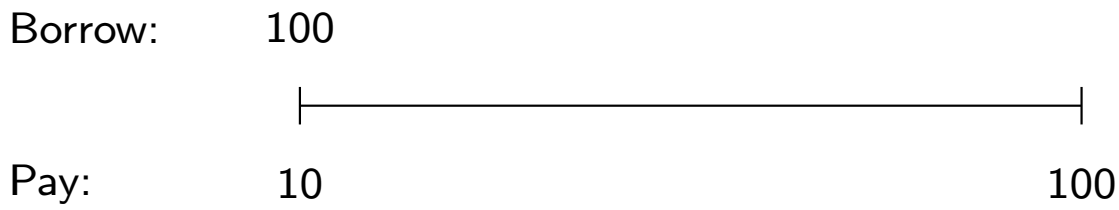
Interest is like rent: it is the fee the borrower pays for the opportunity to use someone else's money.

Suppose you want to borrow \$100 for a year. Bank A offers you a loan at 10% effective rate of interest.



Interest payable up front

Bank B offers you a loan with similar terms but it wants the \$10 interest to be paid up front.



- **Face value**: the amount to be repaid
- **Proceeds**: the amount that the borrower receives
- **Bank discount**: the interest charged up front

Effective rate of discount

Definition

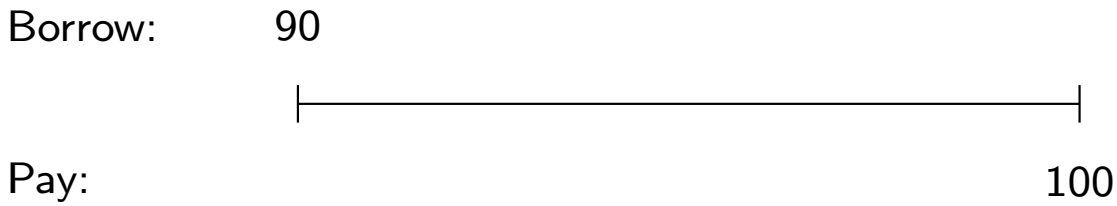
The **effective rate of discount** during the nth period is

$$d_n = \frac{A(n) - A(n-1)}{A(n)} = \frac{I_n}{A(n)}$$

In the example above, the effective rate of discount in Bank B's offer is

Comparing interest and discount

We can think of Bank B's offer as



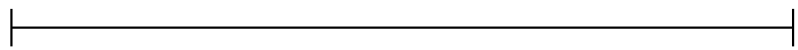
Effective rate of **interest** on this offer is

$$i =$$

Comparing interest and discount

Bank B wants your business and is willing to match Bank A's offer but needs the interest up front. What rate of discount would be equivalent to Bank A's offer (i.e., borrower indifferent between options)?

Borrow:



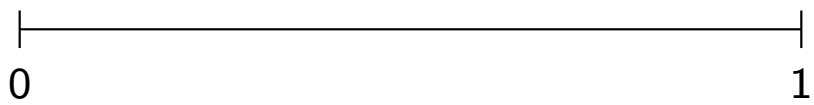
Pay:

Effective rate of discount:

Equivalence of interest & discount

An effective rate of interest of 10% and an effective rate of discount of 9.0909% are equivalent in the sense that

- they produce the same net cash flows
- the interest amounts are the same in present value terms



Equivalence of interest & discount

Mathematically:

$$i = \frac{I}{A(0)} \qquad d = \frac{I}{A(1)} = \frac{i \cdot A(0)}{A(0) + i \cdot A(0)}$$

so

$$\frac{1}{d} = \frac{A(0) + i \cdot A(0)}{i \cdot A(0)} = \frac{1+i}{i} = \frac{1}{iv}$$

Simple discount

Simple discount describes a specific pattern of accumulation:

$$a(t) = \frac{1}{1 - dt}, \qquad t \geq 0$$

where d is the annual rate of simple discount.

Present values are calculated using

$$a(t)^{-1} = 1 - dt, \qquad t \geq 0$$

Simple discount, example

Example 1.6: A bank issues a simple discount note with face value \$4000 due in 3 months. The annual discount rate is 12%. Find the bank interest and the proceeds.

Compound discount

Compound discount is another pattern of accumulation:

$$a(t) = \frac{1}{(1-d)^t}, \qquad t \geq 0$$

where d is the annual rate of compound discount.

Present values are calculated using

$$a(t)^{-1} = (1-d)^t, \qquad t \geq 0$$

Note: using compound discount, the effective rate of discount in each period is constant.

Compound discount, example

Example 1.7: A loan is issued with \$10,000 face value due in 3 years. If the loan is subject to a 12% annual rate of compound discount, find the proceeds to the borrower.

Summary

- Simple and compound discount refer to specific forms of the accumulation function $a(t)$.
- The effective rate of discount is another way of measuring or characterizing the rate at which interest accumulates.

1.7 Nominal Rates

Recall the effective annual rate of interest, which is the % change in the value of an investment from the beginning of the year to the end of the year.

$$i = \frac{I}{A(0)}$$

Example 1.8: Suppose time is measured in quarters. Investment A is such that 100 grows to 105 by the end of the quarter. What is the effective **quarterly** rate of interest?

Comparing rates

Which of the following two options is better for an investor?

- Effective annual rate of interest of 16%, or
- Effective quarterly rate of interest of 4%?

Which of the following two options is better for an investor?

- Effective annual rate of interest of 16%, or
- Effective quarterly rate of interest of 3.9%?

Different compounding periods

How do you compare?

- Pick some time period
- Find the accumulated value at the end of that period under both rates
- Compare accumulated values

Equivalent measures of interest

Measures of interest are said to be **equivalent** if they produce the same accumulated value for a given amount invested for the same length of time.

Example 1.9: What is the effective quarterly interest rate (j) that is equivalent to an effective annual rate of interest of 16%?

Nominal rate of interest

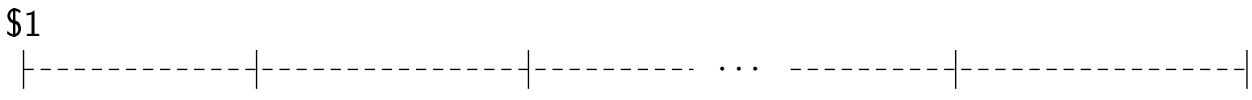
We could also express this pattern of accumulation as a **nominal annual rate of interest**, compounded quarterly:

The measurement period is annual, and "nominal" refers to interest being compounded more often than once per measurement period.

Nominal rate of interest

- The nominal rate of interest payable m times per period is denoted by $i^{(m)}$ where $m = 2, 3, 4, \ldots$
- The rate that applies to each mth of a period is $i^{(m)}/m$.
- The accumulated value after 1 year is $\left(1 + \frac{i^{(m)}}{m}\right)^m$.
- The accumulated value after t years is $\left(1 + \frac{i^{(m)}}{m}\right)^{mt}$.

Accumulation at nominal rate $i^{(m)}$



Time	0	$\frac{1}{m}$	$\frac{2}{m}$		$\frac{m-1}{m}$	1
Interest:		$\frac{i^{(m)}}{m}$	$\frac{i^{(m)}}{m}\left(1+\frac{i^{(m)}}{m}\right)$	$\frac{i^{(m)}}{m}\left(1+\frac{i^{(m)}}{m}\right)^{m-2}$	$\frac{i^{(m)}}{m}\left(1+\frac{i^{(m)}}{m}\right)^{m-1}$	
Balance:	**\$1**	$1+\frac{i^{(m)}}{m}$	$\left(1+\frac{i^{(m)}}{m}\right)^{2}$	$\left(1+\frac{i^{(m)}}{m}\right)^{m-1}$	$\left(1+\frac{i^{(m)}}{m}\right)^{m}$	

Equivalence of nominal rates

A nominal annual rate of $i^{(m)}$ is equivalent to an effective annual rate of i if

$$1 + i = \left(1 + \frac{i^{(m)}}{m}\right)^m .$$

m	1	2	4	12	365
$i^{(m)}$	0.08000	0.07846	0.07771	0.07721	0.07697

As m increases, $i^{(m)}$ decreases. We can prove that as $m \to \infty$, $i^{(m)}$ has a limit. We denote this limit by δ.

Nominal rate of discount

- The nominal rate of discount payable m times per period is denoted by $d^{(m)}$ where $m = 2, 3, 4, \ldots$
- The rate that applies to each mth of a period is $d^{(m)}/m$.
- The accumulated value after 1 year is $1\Big/\left(1 - \frac{d^{(m)}}{m}\right)^m$.
- The accumulated value after t years is $1\Big/\left(1 - \frac{d^{(m)}}{m}\right)^{mt}$.

Equivalence of nominal rates

A nominal annual rate of discount $d^{(m)}$ is equivalent to an effective annual rate of discount d if

$$1 - d = \left(1 - \frac{d^{(m)}}{m}\right)^m .$$

If $i^{(m)}$ and $d^{(m)}$ are equivalent, then

$$\left(1 + \frac{i^{(m)}}{m}\right)^m = 1 \Big/ \left(1 - \frac{d^{(m)}}{m}\right)^m$$

As m increases, $d^{(m)}$ increases. We can prove that as $m \to \infty$, $d^{(m)}$ also converges to δ.

Example 1.10

You want to invest \$100 for a period of 3 years. Bank A offers a nominal rate of interest of 12% convertible quarterly and Bank B offers a nominal rate of discount of 10% convertible semi-annually.

a) Where should you invest your \$100?

Example 1.10, continued

In fact, we could compare accumulated values at any common compounding points:

- After 2 years:

- After 6 months (2 quarters):

Example 1.10, continued

b) What nominal rate of discount convertible twice a year is equivalent to Bank A's offer?